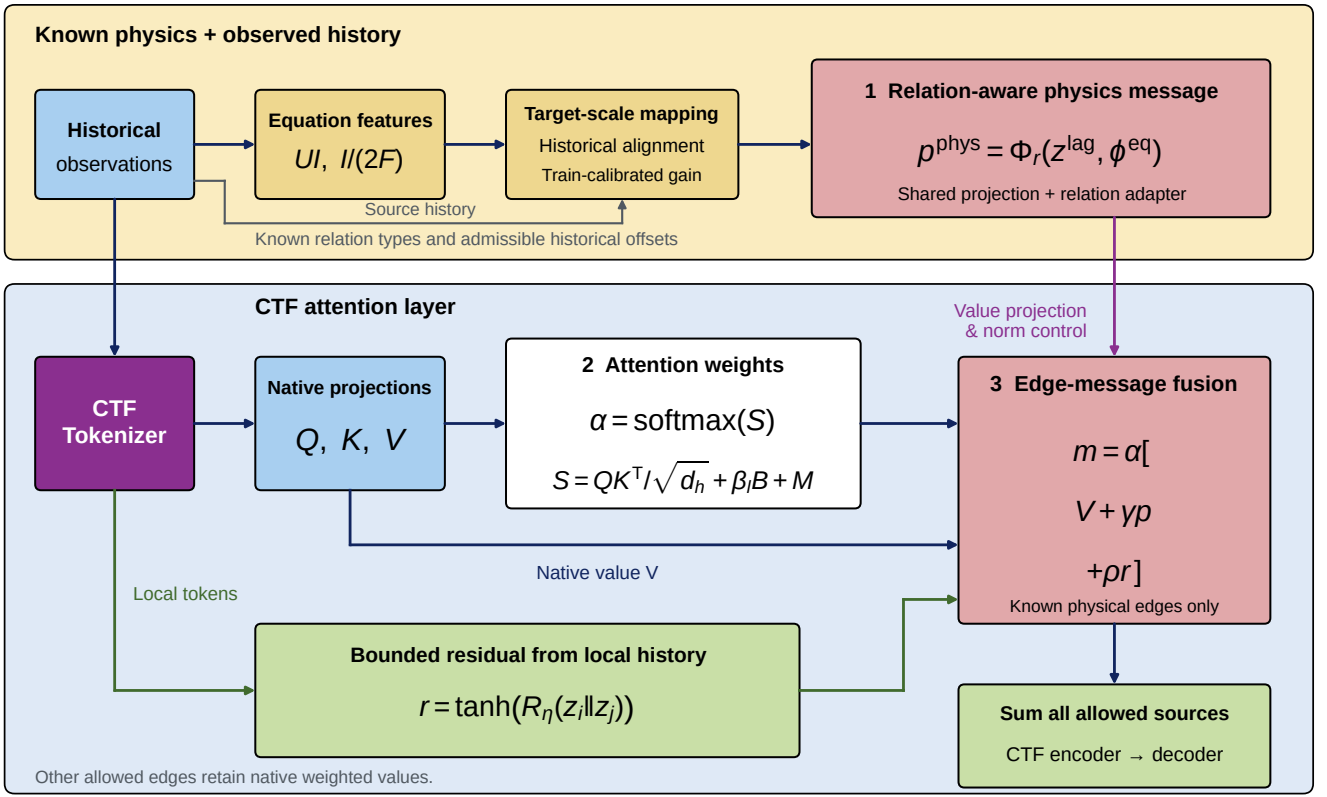
Equation-Derived Messages in CTF
Known physics + observed history
Historical observations
Equation features
UI, II(2F)
Target-scale mapping
Historical alignment
Train-calibrated gain
1 Relation-aware physics message
$p^{phys} = \Phi_r(z^{lag}, \phi^{eq})$
Shared projection + relation adapter
Source history
Known relation types and admissible historical offsets
CTF attention layer
Value projection & norm control
CTF Tokenizer
Native projections
Q, K, V
2 Attention weights
$\alpha = \mathrm{softmax}(S)$
$S = QK^T/\sqrt{d_h} + \beta_l B + M$
3 Edge-message fusion
$m = \alpha[$
$V + yp$
$+\rho r]$
Known physical edges only
Native value V
Local tokens
Bounded residual from local history
$r = \tanh(R_\eta(z_i \| z_j))$
Sum all allowed sources
CTF encoder → decoder
Other allowed edges retain native weighted values.